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Analysis of Optimizing Regional Economic Resource Allocation Efficiency Using Genetic Algorithm (GA)

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***Abstract*—Exploring the application of genetic algorithm (GA) to optimize the efficiency of regional economic resource allocation; Improving the efficiency of resource allocation is crucial in the process of regional economic development. Genetic algorithm is based on the principle of "survival of the fittest" and can provide an effective path for optimizing regional economic resource allocation. Conduct comprehensive research using literature review method, sort out relevant research status, clarify research direction and theoretical basis; Elaborate on the application process of genetic algorithm in regional economic resource allocation through case analysis method; At the same time, using empirical analysis methods, quantitatively evaluate the effectiveness of algorithm optimization in resource allocation efficiency. The research results indicate that by using genetic algorithms to encode regional economic resource elements, evaluating schemes under the influence of fitness functions, and conducting multiple selection, crossover, and mutation operations, the optimal resource allocation scheme can be screened, significantly improving the effectiveness and feasibility of regional economic resource allocation. This demonstrates the practical value and effectiveness of applying genetic algorithms (GA) to optimize the efficiency of regional economic resource allocation, and promotes the research to have practical reference significance. In summary, genetic algorithms can be effectively used for regional economic resource allocation, providing new ideas and methods for improving the quality of regional economic development.**

***Keywords*—Genetic Algorithm (GA), regional economy, resource allocation, efficiency optimization, empirical analysis**

I. INTRODUCTION

Through investigation and research, it has been found that the efficiency of national financial resource allocation has a significant impact on regional economic disparities, both in full-sample and staged examinations. There are differences in the intensity of this impact between northern and southern regions [1]. Meanwhile, only a few provinces in China are considered highly efficient in both stages, with most provinces facing challenges in the smooth transition of resource allocation capabilities between stages [2]. Furthermore, at the level of high-quality development, only cities with a medium level of digital economy demonstrate a promotional effect on regional high-quality development [3]. Although the overall efficiency level of China's technology finance is good, there is still a gap from the efficiency frontier, with the main influencing factor being the rate of technological progress [4]. From the perspective of factor mobility, factor mobility is an important mechanism through which the digital economy improves regional resource allocation efficiency. The improvement effect of the digital economy on resource allocation efficiency is more significant in inland areas than in coastal areas, highlighting the importance of the vigorous development of the digital economy for optimizing resource allocation efficiency [5]. From an international research perspective, some studies have fairly allocated resources to sectors based on certain criteria to achieve optimal outcomes, with these criteria divided into sector strengths and weaknesses [6]. Other studies have integrated a GP with a Social Accounting Matrix (SAM) framework to allocate an economy's economic resources among alternative sectors to achieve several goals [7]. It is evident that how to scientifically achieve effective allocation of regional economic resources is a key research focus in the economic field both domestically and internationally. Based on current research, the integration of modern technology and related algorithms has become the main research direction, presented in data form for qualitative and quantitative analysis. For example, some research has constructed an optimization model for virtual power plant dispatching based on an improved GA, finding optimal dispatching strategies for different units under economic and technical constraints [8]. Another study introduced data asset-related theories and evaluation methods, constructing a GA-BP neural network model to evaluate data assets and quantifying influencing factors from multiple aspects [9]. From the perspective of energy economic resource allocation, GA has been applied to conduct research, intuitively demonstrating the application value and role of the algorithm [10].

Therefore, this study, on the one hand, enriches the research methods of regional economic resource allocation theoretically by introducing GA into this field, providing new ideas for solving complex resource allocation problems. On the other hand, in practice, it provides new support for formulating more scientific and reasonable resource allocation strategies for regions, improving the development quality and competitiveness of the regional economy. At the same time, it deeply analyzes the specific application of GA in regional economic resource allocation, demonstrates it through empirical analysis, and comprehensively considers various resource elements and constraints, making the research results more practical and operable.

II. BASIC CONCEPTS

A. Genetic Algorithm

The GA, first proposed by John Holland from the University of Michigan in the 1970s, is a computational model that simulates the biological evolution process based on Darwin's theory of natural selection and genetic mechanisms. It searches for optimal solutions by mimicking the natural evolution process [11], as illustrated in Fig. 1.

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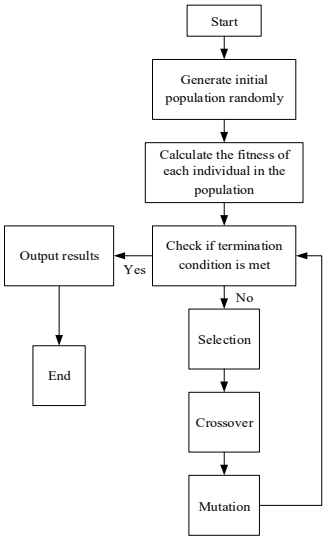


Fig. 1. Basic operating flow of GA.

As shown in Fig. 1, the GA gradually evolves individuals in the population towards the optimal solution through repeated operations of fitness evaluation, selection, crossover, and mutation, until predetermined termination conditions are met, such as reaching the maximum number of generations or no further improvement in fitness values.GA have excellent global search capabilities and do not require strict mathematical assumptions about the problem, making them effective in handling complex multimodal problems. Compared to ant colony algorithm, it converges faster, is less prone to getting stuck in local optima, and has lower computational costs. Compared with gradient descent method, GA does not require differentiable objective function, can handle discrete variables, and has a wider applicability. In the fields of resource allocation, path planning, and machine learning model parameter tuning, GA can find optimal solutions in a more efficient and flexible way.

B. Regional Economic Resource Allocation

Regional economic resource allocation refers to the rational distribution and combination of various economic resources, including human, material, and financial resources, within a specific region to achieve optimal regional economic development [12]. Proper resource allocation ensures full utilization of resources, avoids idleness and waste, enhances production efficiency, and promotes coordinated industrial development. Improper resource allocation may lead to imbalances in the industrial structure, with some industries overdeveloping while others underdeveloping, thereby affecting the sustainable development of the regional economy. Additionally, sustainability is also a crucial factor considered in regional economic resource allocation. By emphasizing environmental protection and ecological balance during the allocation process, harmonious coexistence between the economy and the natural environment can be achieved. For example, the use of clean production technologies can reduce environmental pollution, and the development and utilization of renewable energy can reduce dependence on traditional fossil fuels, enabling green development of the regional economy [13].

III. Optimization Application of GA in Regional Economic Resource Allocation

A. Encoding and Decoding

In applying GA to regional economic resource allocation, it is necessary to encode the resource allocation plans of the regional economy into a chromosomal form that can be processed by GA [14]. This paper adopts real-number encoding, directly using real numbers as chromosomes, which improves the expression accuracy of regional economic resource allocation plans and leverages the advantages of handling continuous variable optimization problems.

Assuming there are multiple (n) resources that need to be allocated within the region, denoted as R_1, R_2, ..., R_n, and the allocation quantity of each resource is denoted as r_1, r_2, ..., r_n, the allocation quantities can be set as a chromosome, i.e., $R = [r_1, r_2, ..., r_n]$. The decoding process converts the chromosome back into the actual resource allocation plan. For real-number encoded chromosomes, the decoding process is relatively straightforward, as it involves directly matching the gene values in the chromosome to the corresponding resource allocation quantities.

B. Fitness Function Design

The fitness function ($F(X_i)$), as a key indicator for measuring the quality of individuals in GA, directly influences the search direction and performance of the algorithm. Therefore, in regional economic resource allocation, the results of an established evaluation system for regional economic resource allocation efficiency can be used as the basic parameters for the fitness function to calculate the efficiency value, which serves as the value of the fitness function. Assuming the efficiency value of the ith individual is U_i, the fitness function can be expressed as:

$$F(X_i) = U_i \tag{1}$$

Where a higher calculated efficiency value indicates a better economic resource allocation plan, corresponding to a higher fitness value [15]. For example, by using the DEA model to evaluate different economic resource allocation plans for a certain region, if the calculated efficiency value for Plan I is 0.84 and for Plan II is 0.77, then in the GA, the fitness value for Plan I is 0.84 and for Plan II is 0.77. Since Plan I has a higher fitness value than Plan II, it is more likely to be selected by the GA for subsequent genetic operations, serving as the gene for forming a better economic resource allocation plan.

C. Implementation of Genetic Operations

Following the GA operations outlined in, the implementation is divided into selection, crossover, and mutation, detailed as follows:

Implementation of Selection Operation: Part of the individuals are selected from the current population to participate in subsequent genetic operations, generating the next generation. The roulette wheel selection method is adopted, where the basic principle is to calculate the selection probability of each individual based on their fitness value. Individuals with higher fitness have a greater probability of being selected. The formula is represented as:

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$$p_i = \frac{F(X_i)}{\sum_{j-1}^{N} F(X_i)} \quad (2)$$

Where N is the number of individuals, and p_i is the selection probability. Through this method, individuals are randomly selected based on probability, with those having higher fitness being more likely to be chosen, thus enabling superior individuals in the population to have more opportunities to reproduce.

Implementation of Crossover Operation: The crossover operation simulates sexual reproduction in the biological world by reconfiguring parts of the chromosomes of selected parent individuals to produce new offspring. The specific steps are as follows:

Step 1: Randomly select two parent individuals from the population, denoted as P_1= [x_{11}, x_{12}, ..., x_{1n}] and P_2= [x_{21}, x_{22}, ..., x_{2n}].

Step 2: Randomly choose a crossover point k, where $k \geq 1$ and $k < n$.

Step 3: For individuals P_1 and P_2, exchange the genes after the crossover point k to generate two new offspring individuals, which inherit some of the excellent characteristics of P_1 and P_2, while also forming new genome combinations. This increases the diversity of the population's genes and provides a basis for subsequent mutation operations, denoted as C_1 and C_2.

Implementation of Mutation Operation: Randomly making small adjustments and changes to an individual's chromosome prevents the algorithm from converging to a local optimum. The basic bit mutation method is employed to ensure that chromosome adjustments and changes occur within a controllable probability, denoted as Pm. For a single gene of an individual, this probability determines whether a mutation operation is performed. If "yes," a new value is randomly generated within the gene's value range to replace the corresponding value. Although the probability of mutation is relatively low, it introduces new genes into the population, extends the algorithm's convergence, reduces the likelihood of reaching a local optimum, and brings the solution closer to the global optimum.

IV. Empirical Analysis

A. Data Sources and Processing

Economic and resource data from a certain region spanning from 2013 to 2023 are used as the analysis sample. The data primarily originates from the region's statistical yearbooks, economic reports published by government departments, and statistical materials from relevant industry associations. The data includes input-output situations of various industries within the region, resource consumption data, population employment information, and economic growth indicators, among others. During data collection, due to differences in data sources, there are variations in data format, statistical standards, and quality. Therefore, data processing is necessary to ensure accuracy, completeness, and consistency. The processing includes the following:

Handling Missing Values: For data with a small number of missing values, methods such as mean imputation, median imputation, or linear interpolation are used to fill in the gaps based on the data's characteristics and distribution. If there are numerous missing values that significantly impact the overall analysis, the corresponding data records and variables are considered for deletion.

Identifying and Handling Outliers: Statistical methods and visualization tools are used to identify outliers in the data. If outliers are due to data entry errors, they are corrected. If they are genuine extreme values, their impact on the results is considered separately in the analysis.

Data Standardization: The Z-Score standardization method is used to standardize the data, represented by the formula:

$$z = \frac{x - \mu}{\sigma} \quad (3)$$

In the formula, x represents the original data; μ denotes the mean of the data; σ stands for the standard deviation of the data. Through standardization, data of different dimensions and magnitudes are transformed into standard data with a mean of 0 and a standard deviation of 1, thereby eliminating the influence of data dimensions on analysis results and making data from different indicators comparable.

B. Result Analysis

By applying the GA, the efficiency of regional economic resource allocation before and after optimization was obtained, allowing for a comparative analysis of the results, as shown in Table I.

TABLE I. RESULTS OF ECONOMIC RESOURCE ALLOCATION EFFICIENCY BEFORE AND AFTER OPTIMIZATION IN A CERTAIN REGION

Evaluation Indicators:	Before Optimization	After Optimization	Change Rate: %
GDP: %	6.5	7.8	19.9
Per Capita GDP: 10,000 Yuan	5.2	5.8	11.5
Industrial Structure Optimization Rate: %	35.0	42.0	20.0
Energy Utilization Rate: %	40.0	45.0	12.5
Land Utilization Rate: %	65.0	70.0	7.7
Unemployment Rate: %	4.8	4.2	-12.5
Resident Income Growth Rate: %	7.0	8.5	21.4

As indicated in Table I, after applying the GA, the efficiency of economic resource allocation in the region significantly improved. Specifically, GDP increased by 19.9%, per capita GDP by 11.5%, the industrial structure optimization rate by 20%, energy utilization rate by 12.5%, land use efficiency by 7.7%, the unemployment rate decreased by 12.5%, and the resident income growth rate increased by 21.4%. Furthermore, multiple experiments revealed that when the population size is 50, the algorithm not only ensures sufficient search capability but also yields good optimization results within an acceptable computation time. As shown in Table II.

Through experimentation, a crossover probability of 0.8 facilitates the evolution of the algorithm towards the optimal solution while maintaining population diversity. With a mutation probability of 1%, new genes can be introduced timely, enhancing the algorithm's global search capability while preserving its stability. Experiments showed that when the maximum number of iterations is 100, the algorithm

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converges within a reasonable time to obtain a superior solution. Beyond this point, increasing the number of iterations yields minimal improvement in optimization results.

TABLE II. POPULATION ITERATION RESULTS

Number of Iterations	1	10	30	50	70	100
The objective function value when the population size is 30, the crossover probability is 0.8, and the mutation probability is 1%	48.6	36.2	20.5	13.8	9.3	5.0
The objective function value when the population size is 50, the crossover probability is 0.8, and the mutation probability is 1%	50.2	38.6	22.3	15.2	10.6	6.0

V. CONCLUSION

In summary, the application of the GA in improving the efficiency of regional economic resource allocation has yielded notable results. Moreover, by adjusting parameters reasonably, it can enhance regional economic growth, optimize the industrial structure, improve resource utilization efficiency, reduce unemployment, and increase resident income. Therefore, in practical applications, a series of experiments and analyses should be conducted based on specific issues and data characteristics of the actual situation to further determine the optimal parameter combination for the GA. This will enhance the advantages and role of the GA in improving the efficiency of regional economic resource allocation, providing a scientific and effective reference for regional economic development decision-making.

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